

NLP Applications in Research and Industry

This course focuses on NLP applications frequently found in industry, such as, e.g., question answering, as well as niche applications with a high potential to improve people's daily life, such as the deployment of NLP models for education or healthcare. The content will be less broad and provide less of an overview than "Intro to NLP" and instead go into more depth on individual applications as well as techniques, such as retrieval-augmented generation (RAG). In addition to applications themselves, the course will also cover aspects relevant to those applications, such as privacy concerns and model reliability (inlacing hallucinations) and interpretability, with an emphasis on how to translate learned topics into code using best practices. This course can be taken either as a follow-up to "Intro to NLP" or independently.

Grading

The university requires passing some percentage/points before being able to be admitted to the final exam which will be the final grade.

The points will be based on:

- Weekly Coding Assignments 40%
- Application Pipeline Projects 40%
- Research papers analysis 20%

Then there will be a final exam

- Final exam 100%

Weekly Coding Assignments — 40%

One coding assignment per week, submitted as a GitHub repository before the start of the following class. Each assignment is scoped to a specific concept or technique from the previous lecture. Each assignment will specify what the code must do, what it must output, and what distinguishes a thorough submission from a minimal one.

Application Pipeline Projects — 40%

One project per application block, due one week after the final lecture of that block.

Each project has two components:

- Report: a written response to the 12-step framework (Problem Definition, Prior Work, Data, Constraints, Architecture, Evaluation, Reliability & Safety, Privacy & Legal, Fairness & Bias, Interpretability, Deployment, Impact). Every sub-question must be addressed. Depth matters more than length.
- Code: a full end-to-end pipeline for the application, including dataset selection (2–3 datasets), evaluation, and training or inference. Will vary depending on the application. Submitted as a GitHub repository.

The report is what distinguishes this from a standard ML project. Students are expected to think through the full lifecycle of the system, not just whether the model runs.

Research Papers Analysis — 20%

Once per application block, four total. For each, you will be given 2–3 papers relevant to that block's application and asked to write a critical analysis. This is not a summary — assume the reader has read the papers. Your analysis should engage with what the authors were trying to do and why, what design choices they made (models, datasets, evaluation), whether those choices were justified, what the results actually show versus what is claimed, and what you would do differently. The goal is to develop the habit of reading research critically rather than taking published results at face value. Strong analyses will draw connections across the provided papers and, where relevant, to topics covered in class.

Rough Schedule

Week 1: Course Introduction

Application 1 — Medical QA (Weeks 2–5)

Week 2: — QA & Language Models

Week 3 — RAG & Hallucination

Week 4 — Medical Domain, Privacy, biases, evaluation, knowledge graphs

Application 2 — Legal Document Retrieval (Weeks 6–8)

Week 5 — Document Understanding

- Research papers analysis due
- Medical QA project due (code + report)

Week 6 — Advanced Retrieval

Week 7 — Translation & Summarization

Application 3 — Educational Tutoring System (Weeks 9–11)

Week 8 — Prompting & Feedback Generation

- Research papers analysis due
- Legal document retrieval project due (code + report)

Week 9 — Evaluation & Interpretability

Week 10 — Bias, Fairness & Human-in-the-Loop

Application 4 — General-Purpose AI Assistants / MoE (Weeks 12–14)

Week 11 — MoE Architecture & Efficiency

- Research papers analysis due
- Educational tutoring system project due (code + report)

Week 12 — MoE Architecture & Efficiency (continue)

Week 13 — Training, Alignment & Safety

Week 14 — Deployment & Wrap-Up

- Research papers analysis due
- General AI project due (code + report)